SUPPORTING ADULT GUIDANCE

Composite shapes and design choices

Perimeter and Area · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Room Splitter, learners choose an L-, T-, U-shaped or stepped room drawn on a 1 m grid. Cut buttons add or remove dashed cut lines; each rectangular piece shows its width × length and area, and the readout adds the pieces. Whatever cuts are chosen, the total floor stays the same. Some walls are marked ?; learners work them out from the walls opposite and can reveal them to check, with the perimeter added up. Every length is a whole number of metres, so every number shown is exact. Six questions and a stretch on why any split gives the same area follow, then on-the-job questions, each with a labelled diagram drawn to scale (flooring packs, costs, centimetres of skirting), true-or-not statements, and a room designer for six design briefs.

Driving Question: *How much floor does this room really have?*

The mathematics, in plain terms

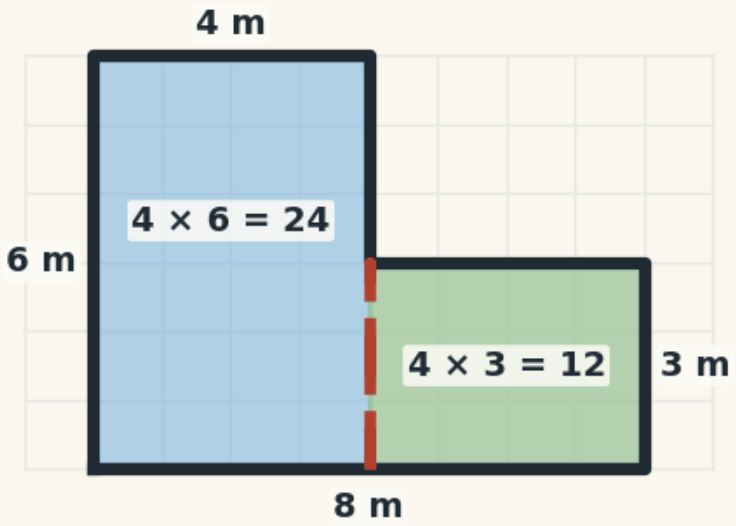
Area is additive: a shape split into non-overlapping pieces has area equal to the sum of the pieces, so every valid split gives the same total. Composite rectilinear areas can also be found by subtraction (bounding rectangle minus the missing parts). For perimeter, the horizontal edges facing one way sum to those facing the other, and the same for vertical edges, which is how missing lengths are found. Removing a rectangular corner from a rectangle leaves the perimeter unchanged (the two new edges replace two equal lengths) but reduces the area; a notch cut into the middle of a side adds twice its depth to the perimeter.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Split an L-shaped room

An L-shaped room is 8 m along one wall and 6 m along another. Find its floor area and its perimeter.

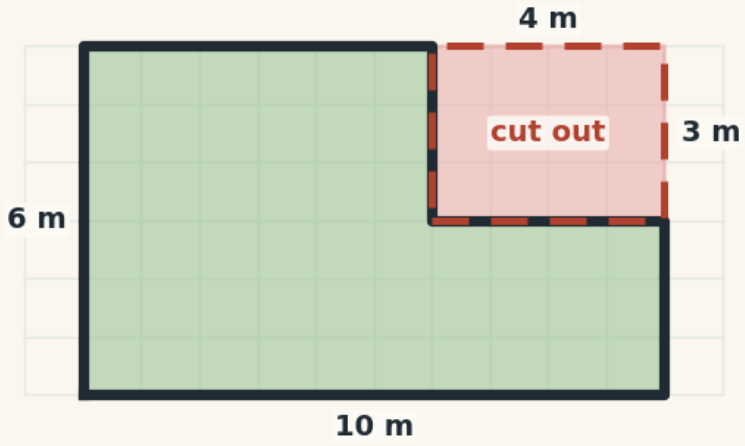


1. Cut down to make two rectangles.	$4 \times 6 = 24$ m^2 and $4 \times 3 = 12$ m^2
2. Add the pieces.	$24 + 12 = 36$ m^2
3. Missing walls: 8 – 4 and 6 – 3	4 m and 3 m
4. Walk round: 8 + 3 + 4 + 3 + 4 + 6	28 m

Floor 36 m^2. Walls 28 m (2800 cm of skirting board).

Example 2 — Take the missing corner away

A 10 m by 6 m room has a 4 m by 3 m corner cut out. Carpet costs £10 per m^2. What does the carpet cost?



1. The whole rectangle $10 \times 6 = 60\ m^2$
2. The missing corner $4 \times 3 = 12\ m^2$
3. The floor $60 - 12 = 48\ m^2$
4. The carpet $48 \times £10 = £480$

Floor 48 m^2; carpet £480.

Split first, then design

Learners split rooms and see the total stay the same before they are asked to calculate, find missing walls, or design. The design briefs then ask them to use area and perimeter together, including a skirting length given in centimetres.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Room Splitter: area by splitting, missing walls and perimeter, with six questions. Stretch: why every split gives the same area.	~16 min
Movement	On the job: areas, walls, centimetres, packs and costs.	~7 min
Reflection	True or not? Area and perimeter of composite rooms.	~7 min
Creativity	The room designer: six design briefs.	~10 min
Rest	However you slice it, the room is the same room.	~2 min

Nutrition	Practical: independence in planning a room.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Multiplying the two longest sides (the whole bounding rectangle).
Ask: “Is all of that rectangle part of the room? Which part is missing?”

Missing a wall, or counting one twice, when finding the perimeter.
Ask: “Start at one corner and walk round. Did you come back to where you started?”

Adding the two known walls when finding a missing one.
Ask: “Together the short walls match the long wall opposite. So what is left?”

Thinking a different split gives a different area.
Ask: “Did the cut add or remove any squares?”

Converting metres to centimetres by multiplying by 10.
Ask: “How many centimetres are in one metre?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: area of rectangles (Lesson 2) and metres and centimetres (Place Value Lesson 8). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says an 8 m by 6 m L-shaped room with a 4 m by 3 m corner missing has a perimeter of 22 m. The aim is a question that brings back the walls.

- “How many walls does the room have?”

- “Can you trace round them with your finger, saying each length?”
- “Which walls did you not have a number for?”
- “What do the two short walls on that side add up to?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Should learners split or subtract?
Either. Encourage them to choose the method with the easier sums and to check by using the other.

Why only whole-metre walls?
So every area and perimeter shown is exact, and learners can count squares to check.

Is the stretch required?
No. The secure core is finding the area of a composite shape by splitting, finding missing walls, and the perimeter.

What comes next?
This completes the Perimeter and Area unit. The unit's Architect's Studio diagnostic can be revisited to see what has moved on.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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